



**COMSATS University Islamabad,
Abbottabad Campus**

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

KOJLO

Version 1.0

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Contents

1.Introduction.....	5
1.1 Purpose.....	5
1.2 Scope	5
2.Overall description.....	5
2.1 Product perspective	5
2.2 Operating environment	6
2.3 Design and implementation constraints	7
3.Requirement identifying technique	7
3.1 Use case description	9
UC-1.....	Error! Bookmark not defined.
UC-2.....	Error! Bookmark not defined.
UC-3.....	Error! Bookmark not defined.
UC-4.....	Error! Bookmark not defined.
UC-5.....	Error! Bookmark not defined.
UC-6.....	Error! Bookmark not defined.
UC-7.....	Error! Bookmark not defined.
UC-8.....	Error! Bookmark not defined.
UC-9.....	Error! Bookmark not defined.
UC-10.....	Error! Bookmark not defined.
UC-11.....	Error! Bookmark not defined.
UC-12.....	Error! Bookmark not defined.
4.Functional Requirements	14
5.Non-Functional Requirements.....	22
6.References	24

Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken

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Signature_____

1.Introduction

This section provides an overview of the Khojlo system and describes the objectives and scope of the Software Requirements Specification document.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Khojlo system. Khojlo is a mobile and web-based business discovery platform designed to help users find newly opened, affordable, and unique local businesses while providing business owners with a cost-effective way to increase their visibility. This document serves as a reference for developers, project supervisors, stakeholders, testers, and future maintainers throughout the development lifecycle.

1.2 Scope

Khojlo is a business discovery and comparison platform that enables users to explore, search, and compare local businesses through a structured and user-friendly interface. The system focuses on promoting newly established and value-driven businesses that often receive limited exposure on existing platforms.

The platform provides features such as user registration and profile management, business registration and management, discovery feeds for new and trending businesses, advanced search and filtering, business comparison, reviews and ratings, map-based location services, AI-powered chatbot assistance, and administrative moderation tools.

The primary goal of Khojlo is to bridge the gap between customers and business owners by improving business visibility, accessibility, and informed decision-making. The system is limited to business discovery and comparison and does not support online ordering, payment processing, or delivery services.

2.Overall description

Here is the overall description of the system.

2.1 Product perspective

Khojlo is a new mobile and web-based business discovery platform developed to address the limitations of existing business listing and recommendation systems. Current platforms such as Google Maps and Yelp primarily promote well-established and highly rated businesses, making it difficult for newly opened and lesser-known businesses to gain visibility.

The proposed system is not a replacement for any existing platform but rather an independent product that complements existing business discovery services by focusing on newly opened, affordable, unique, and value-driven businesses. Khojlo introduces features such as business

comparison, structured discovery feeds, location-based search, and AI-assisted business information retrieval.

2.2 Operating environment

The Khojlo system will operate in the following environment:

OE-1

The system shall operate on Android smartphones running Android 10.0 or later.

OE-2

The system shall operate on modern web browsers including:

Google Chrome (latest versions)

Mozilla Firefox (latest versions)

Microsoft Edge (latest versions)

Apple Safari (latest versions)

OE-3

The frontend application shall be developed using Flutter and Dart and shall support both mobile and web platforms.

OE-4

The backend services shall be hosted on a cloud-based or dedicated server running a Linux operating system.

OE-5

The backend application shall be developed using Python 3.12 and FastAPI.

OE-6

The database shall be implemented using PostgreSQL 18.3 or later.

OE-7

The system shall require a stable internet connection for accessing business information, maps, reviews, and AI chatbot services.

OE-8

The system shall integrate with Google Maps SDK and Google Maps APIs for location tracking, navigation, and map visualization.

OE-9

Users, business owners, and administrators shall access the system remotely through the internet from their respective geographical locations.

2.3 Design and implementation constraints

CO-1

The frontend of the system shall be developed using Flutter 3.24+ and Dart 3.5+ as specified in the project proposal.

CO-2

The backend of the system shall be developed using Python 3.12 and FastAPI.

CO-3

The system shall use PostgreSQL as the primary database management system.

CO-4

The system shall communicate between frontend and backend components through RESTful APIs.

CO-5

The system shall use Google Maps APIs and Google Maps SDK for location-based services and navigation functionality.

CO-6

The AI chatbot module shall use a Retrieval-Augmented Generation (RAG) approach and shall only provide responses based on information stored within the platform database.

CO-7

The project shall follow Object-Oriented Design principles for system development and implementation.

CO-8

The project shall be developed using the Agile Software Development methodology with the Scrum framework.

CO-9

The system shall depend on internet connectivity and shall not support offline operation.

CO-10

The scope of the system shall be limited to business discovery, comparison, reviews, ratings, and information accessibility. Features such as online ordering, payment processing, and delivery services shall not be implemented.

3.Requirement identifying technique

The requirements for the Khojlo system were identified using the Use Case Modeling technique. This technique was selected because Khojlo is an interactive mobile and web-based application that involves direct interaction between different types of users and the system. Use cases help in understanding user goals, system behavior, and the functional requirements needed to achieve those goals. The following Figure 3.1 represents the use case diagram of the Khojlo system.

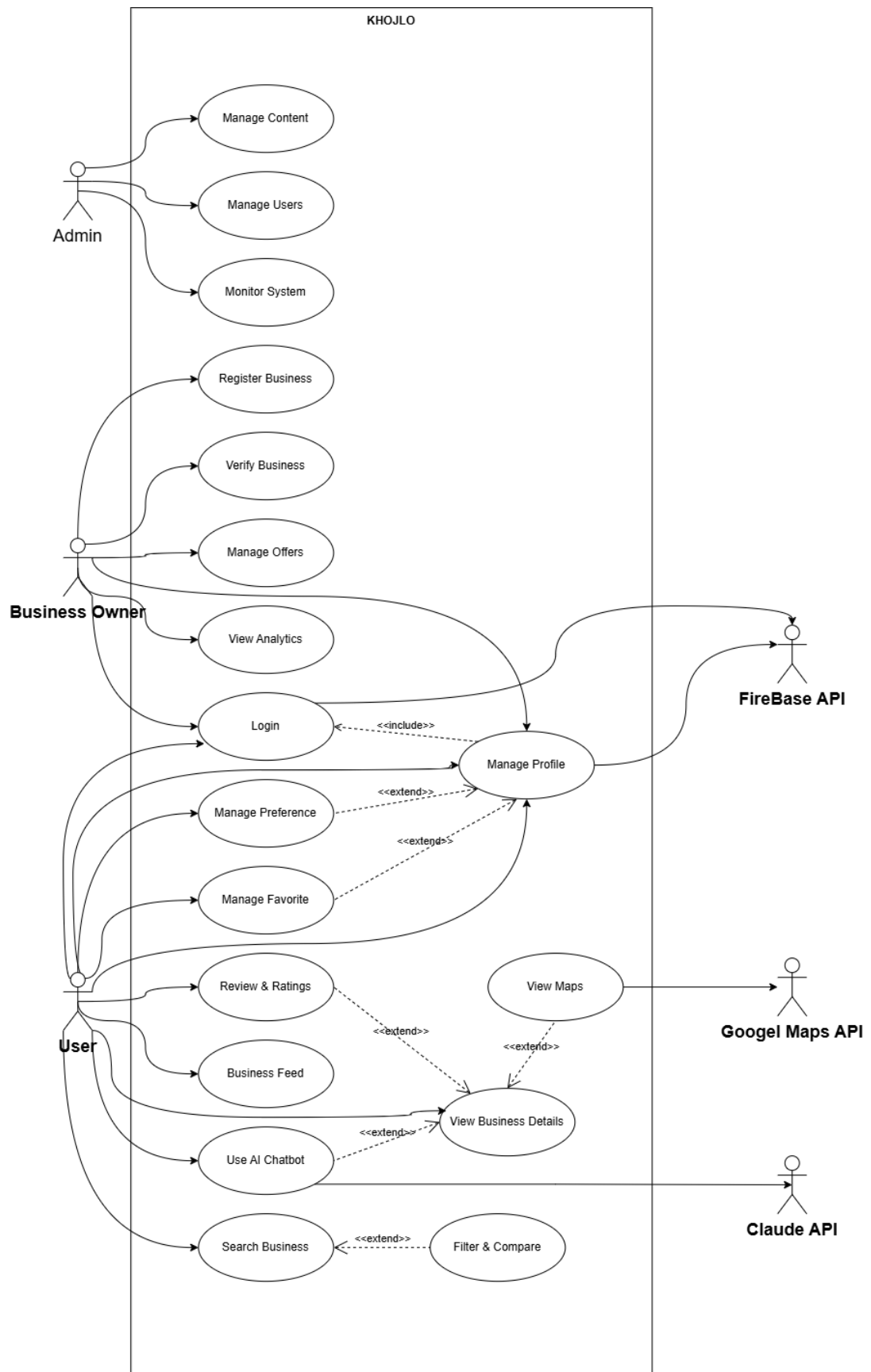


Figure 3.1 Use case diagram of Khojlo

3.1 Use case description

Use cases describe the interactions between the actors and the Khojlo system to achieve specific goals. They capture the functional requirements by defining the actors, triggers, preconditions, normal flow, alternative flows, exceptions, business rules, and postconditions.

Table 3.1.1

Use Case ID	UC-1
Use Case Name	Login
Actors	Customer, Business Owner, Admin
Description	Authenticates registered users.
Trigger	User selects Login.
Preconditions	Account exists.
Postconditions	User authenticated.
Normal Flow	1. Open login page 2. Enter credentials 3. Validate credentials 4. Redirect to dashboard
Alternative Flows	Google login
Exceptions	Invalid credentials
Business Rules	Credentials must match records
Assumptions	Internet available

Table 3.1.2

Use Case ID	UC-2
Use Case Name	Manage Profile
Actors	Customer, Business Owner
Description	Update profile information.
Trigger	Manage Profile selected.
Preconditions	Logged in.
Postconditions	Profile updated.
Normal Flow	1. Open profile 2. Edit details 3. Validate 4. Save
Alternative Flows	Cancel update
Exceptions	Invalid data

Business Rules	Email unique
Assumptions	Authorized user

Table 3.1.3

Use Case ID	UC-3
Use Case Name	Browse Business Feed
Actors	Customer
Description	Browse new and trending businesses.
Trigger	Open feed.
Preconditions	Businesses exist.
Postconditions	Feed displayed.
Normal Flow	1. Open feed 2. Retrieve businesses 3. Display feed
Alternative Flows	Refresh
Exceptions	No businesses
Business Rules	Verified prioritized
Assumptions	Data available

Table 3.1.4

Use Case ID	UC-4
Use Case Name	Search Businesses
Actors	Customer
Description	Search businesses by keyword.
Trigger	Enter search.
Preconditions	Businesses exist.
Postconditions	Results shown.
Normal Flow	1. Enter keyword 2. Search 3. Display results
Alternative Flows	New keyword
Exceptions	No results
Business Rules	Rank by relevance
Assumptions	Internet

Table 3.1.5

Use Case ID	UC-5
Use Case Name	Filter and Compare Businesses
Actors	Customer
Description	Filter and compare businesses.
Trigger	Choose filters.
Preconditions	Search results available.
Postconditions	Filtered comparison shown.
Normal Flow	1. Apply filters 2. Select businesses 3. Compare
Alternative Flows	Clear filters
Exceptions	Comparison unavailable
Business Rules	Max comparison limit
Assumptions	Businesses available

Table 3.1.6

Use Case ID	UC-6
Use Case Name	View Business Details
Actors	Customer
Description	View detailed business information.
Trigger	Select business.
Preconditions	Business exists.
Postconditions	Details displayed.
Normal Flow	1. Select business 2. Retrieve details 3. Display details
Alternative Flows	Open map/reviews
Exceptions	Business unavailable
Business Rules	Only verified businesses
Assumptions	Profile exists

Table 3.1.7

Use Case ID	UC-7
Use Case Name	Submit Reviews and Ratings
Actors	Customer
Description	Submit review.

Trigger	Click Add Review.
Preconditions	Logged in.
Postconditions	Review stored.
Normal Flow	1. Open review 2. Enter rating/comment 3. Submit 4. Save
Alternative Flows	Edit review
Exceptions	Invalid review
Business Rules	One review per business
Assumptions	Customer visited

Table 3.1.8

Use Case ID	UC-8
Use Case Name	Use AI Chatbot
Actors	Customer
Description	Ask business-related questions.
Trigger	Open chatbot.
Preconditions	Knowledge available.
Postconditions	Response displayed.
Normal Flow	1. Ask query 2. Retrieve data 3. Generate response 4. Display
Alternative Flows	Ask another query
Exceptions	No answer
Business Rules	Uses platform data only
Assumptions	AI available

Table 3.1.9

Use Case ID	UC-9
Use Case Name	View Maps
Actors	Customer
Description	View location and directions.
Trigger	Select View Maps.
Preconditions	Location exists.
Postconditions	Map shown.

Normal Flow	1. Open map 2. Load Google Maps 3. Display location
Alternative Flows	Get directions
Exceptions	Location unavailable
Business Rules	Valid coordinates
Assumptions	Maps available

Table 3.1.10

Use Case ID	UC-10
Use Case Name	Register Business
Actors	Business Owner
Description	Register new business.
Trigger	Register selected.
Preconditions	Logged in.
Postconditions	Registration submitted.
Normal Flow	1. Enter details 2. Upload docs 3. Validate 4. Submit
Alternative Flows	Save draft
Exceptions	Missing fields
Business Rules	Unique business
Assumptions	Valid info

Table 3.1.11

Use Case ID	UC-11
Use Case Name	Manage Offers
Actors	Business Owner
Description	Create/update offers.
Trigger	Manage Offers.
Preconditions	Business verified.
Postconditions	Offer saved.
Normal Flow	1. Open offers 2. Add/Edit 3. Save
Alternative Flows	Deactivate offer
Exceptions	Invalid dates

Business Rules	Offer dates valid
Assumptions	Business active

Table 3.1.12

Use Case ID	UC-12
Use Case Name	Verify Business
Actors	Admin
Description	Verify submitted businesses.
Trigger	Pending request.
Preconditions	Admin logged in.
Postconditions	Status updated.
Normal Flow	1. Review request 2. Validate 3. Approve/Reject
Alternative Flows	Request more info
Exceptions	Verification failed
Business Rules	Only admin verifies
Assumptions	Documents provided

4.Functional Requirements

The functional requirements describe the services and functions that the Khojlo system shall provide to its users. These requirements are organized according to the major features of the system.

4.1 User Authentication and Profile Management

This feature enables users to register, log in, and manage their profiles.

Table 4.1.1

Identifier	FR-1
Title	User Registration
Requirement	The user shall be able to create a new account using email and password.

Source	User
Rationale	Allow new users to access the platform.
Business Rule	BR-1 Email must be unique
Dependencies	None
Priority	High

Table 4.1.2

Identifier	FR-2
Title	User Login
Requirement	The user shall be able to log into the system using valid credentials.
Source	User
Rationale	Provide secure access to platform services.
Business Rule	BR-2 Credentials must be validated
Dependencies	FR-1
Priority	High

4.2 Business Discovery and Search

Users can discover, search, filter and view detailed information about businesses.

Table 4.2.1

Identifier	FR-3
Title	Search Businesses

Requirement	The user shall be able to search businesses using keywords and categories.
Source	User
Rationale	Enable business discovery.
Business Rule	None
Dependencies	FR-2
Priority	High

Table 4.2.2

Identifier	FR-4
Title	Filter Businesses
Requirement	The user shall be able to filter businesses by category, location and price.
Source	User
Rationale	Improve search accuracy.
Business Rule	None
Dependencies	FR-3
Priority	High

Table 4.2.3

Identifier	FR-5
Title	View Business Details

Requirement	The system shall display complete business information.
Source	User
Rationale	Provide informed decision making.
Business Rule	BR-3 Only verified business profiles displayed
Dependencies	FR-3
Priority	High

4.3 User Reviews and Favorites

Users can leave reviews, submit ratings, and save favorite businesses for quick access.

Table 4.3.1

Identifier	FR-6
Title	Submit Reviews
Requirement	The user shall be able to submit ratings and reviews.
Source	User
Rationale	Support transparency and trust.
Business Rule	BR-4 One review per business per user
Dependencies	FR-2
Priority	High

Table 4.3.2

Identifier	FR-7
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Title	Save Favorites
Requirement	The user shall be able to save businesses to favorites.
Source	User
Rationale	Allow quick future access.
Business Rule	None
Dependencies	FR-2
Priority	Medium

4.4 Business Owner Management

Business owners can register their businesses, manage profiles, and create promotional offers.

Table 4.4.1

Identifier	FR-8
Title	Business Registration
Requirement	Business owners shall be able to register businesses.
Source	Business Owner
Rationale	Allow businesses to join platform.
Business Rule	BR-5 Business verification required
Dependencies	FR-2
Priority	High

Table 4.4.2

Identifier	FR-9
Title	Manage Business Profile
Requirement	Business owners shall be update business information.
Source	Business Owner
Rationale	Maintain accurate information.
Business Rule	BR-5 Business verification required
Dependencies	FR-8
Priority	High

Table 4.4.3

Identifier	FR-10
Title	Manage Promotions
Requirement	Business owners shall create promotional offers.
Source	Business Owner
Rationale	Increase business visibility.
Business Rule	None
Dependencies	FR-9
Priority	Medium

4.5 System Features

Core system features including discovery feed, map integration, and AI chatbot support.

Table 4.5.1

Identifier	FR-11
Title	Discovery Feed
Requirement	The system shall display new and trending businesses.
Source	System
Rationale	Promote new businesses.
Business Rule	BR-6 New businesses prioritized
Dependencies	FR-8
Priority	High

Table 4.5.2

Identifier	FR-12
Title	Map Integration
Requirement	The system shall display business locations on maps.
Source	User
Rationale	Assist navigation.
Business Rule	BR-7 Valid GPS coordinates required
Dependencies	FR-5
Priority	High

Identifier	FR-13
Title	AI Chatbot
Requirement	The system shall answer business-related queries using RAG.
Source	User
Rationale	Provide instant assistance.
Business Rule	BR-8 Responses based on stored platform data
Dependencies	FR-5
Priority	High

4.6 Admin and Moderation

Administrative functions for verifying business profiles and moderating platform content.

Table 4.6.1

Identifier	FR-14
Title	Admin Verification
Requirement	Admin shall verify business profiles before publication.
Source	Admin
Rationale	Ensure authenticity.
Business Rule	BR-9 Only admins may verify businesses
Dependencies	FR-8
Priority	High

Table 4.6.2

Identifier	FR-15
Title	Content Moderation
Requirement	Admin shall remove spam and inappropriate content.
Source	Admin
Rationale	Maintain platform quality.
Business Rule	BR-10 Moderation policies enforced
Dependencies	FR-14
Priority	High

5.Non-Functional Requirements

This section describes the quality attributes and operational characteristics that the Khojlo system must satisfy to ensure a reliable, efficient, secure, and user-friendly experience.

5.1 Usability

The usability requirements ensure that users can easily learn and interact with the Khojlo system.

USE-1: The system shall allow users to search for businesses using a maximum of three interactions from the home screen.

USE-2: The system shall provide a consistent user interface across mobile and web platforms.

USE-3: The system shall display clear validation messages whenever users enter invalid data.

USE-4: The system shall allow users to access business details directly from search results with a single click/tap.

USE-5: The AI chatbot interface shall always be accessible from the main navigation menu.

USE-6: New users shall be able to complete account registration within 2 minutes under normal conditions.

5.2 Performance

The performance requirements define the expected speed and responsiveness of the system.

PER-1: 95% of application pages shall load completely within 4 seconds over a 20 Mbps or faster internet connection.

PER-2: The system shall return search results within 3 seconds for 95% of search requests.

PER-3: The system shall display business profile details within 2 seconds after user selection.

PER-4: The AI chatbot shall respond to user queries within 5 seconds for 90% of requests.

PER-5: The system shall support at least 200 concurrent users without significant degradation in performance.

5.3 Reliability

The system shall provide dependable and consistent operation.

REL-1: The system shall maintain data integrity during all create, update, and delete operations.

REL-2: The system shall automatically recover from temporary server failures without loss of stored data.

REL-3: The system shall successfully process at least 99% of valid user requests.

REL-4: User reviews, ratings, and business information shall remain persistent after system restarts.

5.4 Security

The system shall protect user data and prevent unauthorized access.

SEC-1: All user passwords shall be stored in encrypted form.

SEC-2: Only verified business owners shall be allowed to modify business profiles.

SEC-3: Administrative functions shall be accessible only to authorized administrators.

SEC-4: The system shall use secure HTTPS communication for all client-server interactions.

5.5 Scalability

The system shall be capable of handling future growth.

SCA-1: The system architecture shall support the addition of new business categories without requiring major modifications.

SCA-2: The system shall support future expansion to additional cities and geographical regions.

SCA-3: The database shall support growth in the number of users, businesses, reviews, and ratings without significant performance degradation.

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